The Fermionic Matter Sub-Programme

Presentation Note 6

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Abstract

The fermionic matter sub-programme of the Cosmochrony corpus addresses a single central question: where do fermions, chirality, hypercharge, and three generations come from? The answer is that they are not postulated as external fields: they arise as the spinorial face of the admissible Weil module V_ρ, once its metaplectic Lie-algebraic structure is complexified by the Lorentzian metric established in the geometric branch. The structural chain is:

$$\Pi \Rightarrow F_n \simeq V_\rho \Rightarrow \mathrm{Mp}(2,\mathbb{R}) \Rightarrow \mathrm{mp}(2,\mathbb{R})_{\mathbb{C}} \simeq \mathfrak{sl}_2(\mathbb{C}) \rightsquigarrow \mathrm{Spin}(3,1) \simeq \mathrm{SL}(2,\mathbb{C}) \Rightarrow \mathcal{S}_\Pi.$$

Three modular structural results are established in Q14: (A) the admissible spinor bundle $\mathcal{S}_\Pi$ reproduces the $\mathrm{SU}(2)_L \times \mathrm{U}(1)_Y$ electroweak bundle structure from tensor functors of $\mathcal{S}_\Pi$; (B) the projected Dirac operator $D_{\Pi,g,A}$ contains a canonical projective endomorphism E_Π that enforces the $V - A$ chiral structure (conditional on Hypothesis 3.5) and whose γ_5-weighted a_4 coefficient imposes anomaly-cancellation constraints making hypercharge a spectral datum; (C) the saturation invariant $\sigma_c(n_3) = 3$ has a spinorial multiplicity reading yielding a gauge-singlet three-generation factor C^3_{gen}. The colour-coupled quark sector is conditional on [H-color]. Open deliverables are the explicit form of E_Π, the Yukawa sector, and the derivation of Hypothesis 3.5 from A1–A4.

Keywords. fermionic matter; admissible spinor bundle; metaplectic lift; chirality; V-A structure; hypercharge rigidity; anomaly cancellation; three generations; projected Dirac operator; Weil representation; non-injective projection; presentation note.

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1 Motivation and Central Question

The fermionic matter sub-programme answers the following question.

Central question. The bosonic spectral stratification — gravity at a_2, Yang–Mills at a_4 — derives the bosonic sector of the Standard Model from the same admissible spectral functional. Fermions, chirality, hypercharge assignments, and the three-generation structure are not covered by the bosonic synthesis. *Do these fermionic structures also arise as forced consequences of the admissible Weil fibre, or must they be introduced as additional postulates?*

The answer established in Q14 [1] is: they are forced. Fermions arise as the spinorial face of the Weil module V_ρ after Lorentzian complexification of its metaplectic lift. Chirality and the $V - A$ structure arise from the projective endomorphism E_Π of the projected Dirac operator, under Hypothesis 3.5. Hypercharge is not a free parameter: it is constrained by spectral anomaly cancellation. Three generations arise from the spinorial multiplicity reading of the same rank-three invariant $\sigma_c(n_3) = 3$ that yields the three spatial directions in the geometric branch.

This sub-programme is the first paper of the Q-series that depends simultaneously on all four layers of the programme: the admissible Weil fibre structure (Notes 1 and 5), the closed Lorentzian geometry (Note 2), the gauge bundle (Note 3), and the gauge–gravity synthesis (Q13 [2], to be summarised in Presentation Note 8).

Remark 1 (Scope: identification only, not mass spectrum)**.** This sub-programme closes the *identification* of the fermionic sector: spinor bundle, chirality, hypercharge quantum numbers, and generation factor. It does not derive the mass spectrum, the Yukawa couplings, or the CKM/PMNS mixing matrices. Those require the explicit construction of E_Π as an operator on the spectral space, and the integration of the projected Yukawa Lagrangian $\mathcal{L}_Y$, both of which are listed as open deliverables.

Remark 2 (Vocabulary note)**.** Throughout this note, *admissibility constraint* replaces the earlier "relaxation" vocabulary. The substrate χ is static; fermions are not excitations of χ but the spinorial interpretation of the admissible fibre after geometric complexification.

2 Position in the Cosmochrony Programme

The fermionic matter sub-programme sits at the apex of Branch III. It depends on all five preceding sub-programmes and provides the fermionic matter content consumed by any downstream phenomenological application.

Inputs from upstream:

- $F_n \simeq V_\rho \simeq L^2(\mathbb{Z}/q\mathbb{Z})$, canonical symplectic structure of V_ρ, and the parity involution $c \leftrightarrow q - c$ (Notes 1 and 5).
- Effective Lorentzian metric $g^{\mu\nu} = 2\eta^{\mu\nu}$ and effective base manifold $M_{\text{eff}} = \mathbb{R}_\tau \times \text{Heis}_3(\mathbb{R})$ (Note 2, Q5b–Q11).
- Admissible principal bundle $P_{G_\Pi}(M_{\text{eff}}, G_\Pi)$ and gauge connection A (Note 3, Q6a).
- $\text{SU}(3) \times \text{SU}(2) \times \text{U}(1)$ gauge group (Note 3).
- Spectral entropy functional $S_\Pi[g, A]$ and the a_4 variation (Note 4/Q12, Q13 [2]).
- $H_{\text{eff}} = \text{Sym}^2(V_\rho)$, three spatial directions from $\operatorname{Im}\mathbb{H}$, $\sigma_c(n_3) = 3$ (Note 2, Q7 [3]).

3 Logical Chain of the Sub-Programme

The sub-programme is organised around the following structural chain.

$$\Pi \Rightarrow F_n \simeq V_\rho \;\Rightarrow\; \mathrm{Mp}(2,\mathbb{R}) \;\Rightarrow\; \mathrm{mp}(2,\mathbb{R})_{\mathbb{C}} \simeq \mathfrak{sl}_2(\mathbb{C}) \;\rightsquigarrow\; \mathrm{Spin}(3,1) \simeq \mathrm{SL}(2,\mathbb{C}) \;\Rightarrow\; \mathcal{S}_\Pi \tag{1}$$
$$\Rightarrow \underbrace{\mathrm{Sym}^2(\mathcal{S}_\Pi) \Rightarrow \mathrm{ad}_{\mathbb{C}}(P_\Pi)}_{\mathrm{SU}(2)_L},\; \underbrace{\wedge^2(\mathcal{S}_\Pi) = L_Y}_{\mathrm{U}(1)_Y},\; \underbrace{E_\Pi \text{ left-admissible}}_{V-A},\; \underbrace{\sigma_c(n_3) = 3 \to C^3_{\mathrm{gen}}}_{3 \text{ generations}}.$$

Three distinct mechanisms contribute.

1. *Lorentzian complexification forces the spin group.* The Weil module V_ρ carries a canonical symplectic structure; its automorphisms lift to $\mathrm{Mp}(2,\mathbb{R})$. The effective metric $g^{\mu\nu} = 2\eta^{\mu\nu}$ (Lorentzian) forces the Lie algebra complexification $\mathrm{mp}(2,\mathbb{R})_{\mathbb{C}} \simeq \mathfrak{sl}_2(\mathbb{C})$, whose Lorentzian integration is $\mathrm{Spin}(3,1) \simeq \mathrm{SL}(2,\mathbb{C})$. The spinor bundle $\mathcal{S}_\Pi$ is the associated fundamental bundle.

2. *Tensor functors of $\mathcal{S}_\Pi$ reproduce the electroweak bundle.* $\mathrm{Sym}^2(\mathbb{C}^2) \simeq \mathfrak{sl}_2(\mathbb{C})$: the symmetric sector gives the complex adjoint, from which the compact real form selects the $\mathrm{SU}(2)_L$ gauge sector. $\wedge^2(\mathbb{C}^2) \simeq \det(\mathbb{C}^2)$: the determinant line L_Y supports the $\mathrm{U}(1)_Y$ factor. No independent electroweak matter bundle is introduced.

3. *The projective endomorphism E_Π enforces chirality and hypercharge.* The projected Dirac operator satisfies the Lichnerowicz-type identity

$$D^2_{\Pi,g,A} = -(\nabla^{S,A})^2 + \frac{R}{4} + \frac{1}{2}\gamma^\mu\gamma^\nu F^a_{\mu\nu} T^a_R + E_\Pi.$$

   Under Hypothesis 3.5 (spinorial lift of BI parity), E_Π is left-admissible ($P_R E_\Pi P_R = 0$), giving the $V - A$ structure. $\mathrm{U}(1)_Y$ invariance of $S_\Pi[g, A, D_{\Pi,g,A}]$ imposes the anomaly-cancellation trace constraint $\mathrm{Tr}_{\mathcal{S}_\Pi}(\gamma_5 Y A_\Pi(x)) = 0$, fixing hypercharge as a spectral coherence datum.

4 Constituent Papers

The sub-programme has a single primary paper.

4.1 Q14: Fermionic matter and chirality from projective Dirac admissibility

Q14 [1] is the sole constituent paper. It is organised around three independent but mutually compatible structural results.

Theorem A: Spinorial electroweak bundle (structural; unconditional). The admissible Weil module V_ρ induces, via the metaplectic lift and the Lorentzian complexification forced by $g^{\mu\nu} = 2\eta^{\mu\nu}$, an admissible spinor bundle $\mathcal{S}_\Pi$. Its tensor sectors satisfy:

$$\mathrm{Sym}^2(\mathcal{S}_\Pi) \simeq \mathrm{ad}_{\mathbb{C}}(P_\Pi), \qquad \wedge^2(\mathcal{S}_\Pi) = L_Y.$$

After selection of the compact real form, these define the $\mathrm{SU}(2)_L \times \mathrm{U}(1)_Y$ electroweak bundle data without any additional input.

Theorem B: Chiral selection and hypercharge rigidity (structural; conditional on Hypothesis 3.5). The projected Dirac operator $D_{\Pi,g,A}$ has the Lichnerowicz-type square above. Under Hypothesis 3.5 (the spinorial lift of BI parity maps right-chiral modes to non-admissible directions), E_Π is left-admissible: $P_R E_\Pi P_R = 0$. This is the $V - A$ structure. $\mathrm{U}(1)_Y$ invariance of the projected spectral functional then imposes, at the a_4 level:

$$\mathrm{Tr}_{\mathcal{S}_\Pi}(\gamma_5 Y A_\Pi(x)) = 0 \quad \forall x \in M_{\mathrm{eff}},$$

where A_Π is the γ_5-weighted a_4 Seeley–DeWitt density of $D^2_{\Pi,g,A}$. This constraint is not postulated: it is the chiral consistency condition of the same a_4 level that yields Yang–Mills dynamics in the bosonic variation. The hypercharge weights Y_R are thereby not free parameters but spectral coherence data.

Theorem C: Generation multiplicity (structural; unconditional). The saturation invariant $\sigma_c(n_3) = 3$ has two functorially distinct readings of the same rank-three admissible invariant. In the geometric branch (Note 2), it yields three effective spatial directions. In the spinorial branch, the spinorial multiplicity functor maps it to a gauge-singlet generation space:

$$C^3_{\text{gen}} \subset \ker(\text{ad}_{\text{SU}(2)} \oplus Y).$$

The identity $\mathbb{R}^3_{\text{space}} \neq C^3_{\text{gen}}$ is established explicitly (Q14 §5): the spatial and generation spaces are distinct functorial images of the same admissible invariant. Three generations are forced; their physical distinction requires the explicit construction of E_Π (open).

Colour sector (conditional on [H-color]). The colour-coupled quark sector $\mathcal{S}_\Pi \otimes V_{\text{color}}$ is obtained by tensoring the admissible spinor bundle with the colour module V_{color}. The status of this sector inherits the status of [H-color] from Notes 1 and 3: established unconditionally on the colour-adapted graph, and supported on the standard graph at the rank, averaged, effective-exponent, and numerical levels. The remaining open component is $[\text{H-color}]_{\text{pointwise}}$.

Conditional status and Hypothesis 3.5. Hypothesis 3.5 is the spinorial lift of BI parity to the spinor bundle $\mathcal{S}_\Pi$. It asserts that the BI parity involution $c \leftrightarrow q - c$ — proved at the scalar Weil level in O18 and BornInfeld — extends to an anti-linear map on $\mathcal{S}_\Pi$ that maps right-chiral modes to non-admissible directions. This hypothesis is structurally motivated by the BI parity of the scalar sector, but is not derived from A1–A4 in the current paper. All uses of chirality and $V - A$ in Q14 depend explicitly on Hypothesis 3.5.

Table 1: Summary of Q14 results by theorem. Status: P = proved; S = structural; C = conditional.

Result	Central output	Status
Theorem A	Spinor bundle $\mathcal{S}_\Pi$; EW bundle from tensor functors	S (uncond.)
Theorem B ($V - A$)	$P_R E_\Pi P_R = 0$; $V - A$ structure	S (cond. on Hyp. 3.5)
Theorem B (hypercharge)	Y_R from anomaly-cancellation trace; not free	S (cond. on Hyp. 3.5)
Theorem C	C^3_{gen}; three generations as gauge-singlet factor	S (uncond.)
Colour sector	$\mathcal{S}_\Pi \otimes V_{\text{color}}$; quark bundle	S (cond. on H-color)

5 Inputs from Upstream Results

1. $F_n \simeq V_\rho$ **and symplectic structure** (Notes 1, 5): the identification of the admissible fibre as the Weil representation is the starting point of the metaplectic lift. Without this, the spinor bundle $\mathcal{S}_\Pi$ has no canonical definition.

2. **Lorentzian metric** $g^{\mu\nu} = 2\eta^{\mu\nu}$ (Note 2, Q5b–Q11): the Lorentzian signature forces the complexification $\text{mp}(2,\mathbb{R})_\mathbb{C} \simeq \mathfrak{sl}_2(\mathbb{C})$. Without the Lorentzian metric, the spin group would remain $\text{Sp}(2,\mathbb{R})$ and no Dirac spinors would arise.

3. **Gauge bundle and** $\text{SU}(2)_L \times \text{U}(1)_Y \times \text{SU}(3)$ (Note 3, Q6a): the admissible principal bundle and gauge connection A enter the spin–gauge connection $\nabla^{S,A}$ of the projected Dirac operator.

4. a_4 **Yang–Mills variation** (Q12 [4], Q13 [2]): the spectral anomaly cancellation of Theorem B uses the same a_4-level structure that yields Yang–Mills dynamics in the bosonic sector. The fermionic a_4 coefficient is a structural extension of the bosonic one. The gauge–gravity synthesis of Q13 is the scientific dependency here; its presentation-level treatment will appear in Presentation Note 8.

5. $\sigma_c(n_3) = 3$ (Note 1, O23): the rank-three saturation invariant provides the spinorial multiplicity input for Theorem C.

6. **BI parity involution** (BornInfeld, O18): the scalar BI parity $c \leftrightarrow q - c$ is the structural motivation for Hypothesis 3.5; its derivation at the scalar level is complete.

6 Outputs to Downstream Branches

Table 2: Principal outputs and downstream consumers.

Output	Consumer	Status
Admissible spinor bundle $\mathcal{S}_\Pi$	Phenomenology, Q-series	S
$\mathrm{SU}(2)_L \times \mathrm{U}(1)_Y$ from tensor functors	SM phenomenology	S
$V - A$ chiral structure	Weak interaction pheno	S (Hyp. 3.5)
Hypercharge as spectral datum	SM charge assignment	S (Hyp. 3.5)
Three-generation factor C^3_{gen}	Mass/mixing notes	S
Quark bundle $\mathcal{S}_\Pi \otimes V_{\mathrm{color}}$	QCD sector	S (H-color)
E_Π as projective endomorphism	Yukawa sector (open)	S (open)

7 Status of Results

7.1 Structural results (unconditional)

1. *Admissible spinor bundle* (Theorem A): $\mathcal{S}_\Pi$ is induced from V_ρ via the metaplectic lift and Lorentzian complexification. The construction is algebraic and functorial; it depends on the Lorentzian closure of Q5b–Q11 but not on Hypothesis 3.5.

2. $\mathrm{SU}(2)_L$ *from* $\mathrm{Sym}^2(\mathcal{S}_\Pi)$ (Theorem A): the symmetric tensor sector gives the complex adjoint $\mathrm{ad}_{\mathbb{C}}(P_\Pi) \simeq \mathfrak{sl}_2(\mathbb{C})$; after compact real-form selection, the $\mathrm{SU}(2)_L$ gauge sector is identified.

3. $\mathrm{U}(1)_Y$ *from* $\wedge^2(\mathcal{S}_\Pi)$ (Theorem A): the determinant line $L_Y = \wedge^2(\mathcal{S}_\Pi)$ is the geometric support of the hypercharge factor.

4. *Generation multiplicity* (Theorem C): $C^3_{\mathrm{gen}} \subset \ker(\mathrm{ad}_{\mathrm{SU}(2)} \oplus Y)$ is a gauge-singlet three-generation space arising from the spinorial multiplicity reading of $\sigma_c(n_3) = 3$. The spatial and generation readings of the rank-three invariant are distinct ($\mathbb{R}^3_{\mathrm{space}} \neq C^3_{\mathrm{gen}}$; Q14 §5).

7.2 Structural results (conditional on Hypothesis 3.5)

1. *Left-admissibility of* E_Π (Theorem B): $P_R E_\Pi P_R = 0$ follows from Hypothesis 3.5 applied to the spinorial sector.

2. *$V - A$ chiral structure* (Theorem B): the condition $P_R E_\Pi P_R = 0$ selects the left-chiral admissible projection $P_L \mathcal{S}_\Pi$, giving the $V - A$ coupling of weak interactions.

3. *Hypercharge rigidity* (Theorem B): the anomaly-cancellation trace condition $\mathrm{Tr}_{\mathcal{S}_\Pi}(\gamma_5 Y A_\Pi(x)) = 0$ constrains Y_R to spectral coherence data; hypercharge is not a free parameter of the framework.

4. *Quark bundle* (§6): $\mathcal{S}_\Pi \otimes V_{\mathrm{color}}$ is structural conditional on both Hypothesis 3.5 and [H-color].

7.3 Open hypotheses

1. *Hypothesis 3.5: spinorial lift of BI parity.* The BI parity involution $c \leftrightarrow q - c$ is proved at the scalar Weil level (O18, BornInfeld). Its extension to an anti-linear endomorphism of $\mathcal{S}_\Pi$ mapping right-chiral modes to non-admissible directions is Hypothesis 3.5: structurally motivated, not yet derived from A1–A4. This is the sole remaining conditionality for the $V - A$ structure and hypercharge rigidity.

2. *Explicit form of E_Π.* E_Π is identified as the zero-order endomorphism of the projected Dirac square, the "internal spectral residue of non-injective projection." Its explicit form as an operator on the spectral space — analogous to the finite internal Dirac operator of Connes–Chamseddine spectral triples — is not yet constructed. Without the explicit E_Π, the three-generation distinction and the Yukawa sector cannot be derived.

3. *Normalisation of $\mathcal{L}_Y$.* The Yukawa Lagrangian $\mathcal{L}_Y$ is identified as arising from the trace $\mathrm{Tr}_{\mathcal{S}_\Pi}[E_\Pi^2]$ variation. Its integral normalisation and the derivation of the mass matrix from the spectral data of E_Π are open.

4. $[\text{H-color}]_{\text{pointwise}}$. Inherited from Notes 1 and 3. The quark bundle $\mathcal{S}_\Pi \otimes V_{\mathrm{color}}$ is fully established once this is resolved.

8 Open Deliverables

Three open deliverables define the boundary of the sub-programme.

Open deliverable 1: Derivation of Hypothesis 3.5 from A1–A4. Hypothesis 3.5 is the only remaining non-axiomatic input to the fermionic sector after O18 and BornInfeld are accepted. Its derivation from the admissibility axioms would make Theorems B and C unconditional. The structural mechanism is identified (BI parity at the scalar level is the model); the spinorial extension is an open algebraic problem in the representation theory of $\mathrm{Mp}(2, \mathbb{R})$.

Open deliverable 2: Explicit E_Π and the Yukawa sector. The mass spectrum of charged leptons and quarks, and the CKM/PMNS mixing matrices, require the explicit spectral construction of E_Π as an operator on the admissible Hilbert space. This requires integrating the projected Yukawa Lagrangian $\mathcal{L}_Y$ and extracting the mass matrix from the eigenvalues of E_Π^2. This is the primary open problem of the fermionic sector.

Open deliverable 3: Full matter content in Lorentzian signature. The complete fermionic content $S_\Pi^{\mathrm{matter}} = (\mathcal{S}_\Pi \oplus (\mathcal{S}_\Pi \otimes V_{\mathrm{color}})) \otimes C^3_{\mathrm{gen}}$ in full Lorentzian signature, with explicit gauge couplings and three-generation Yukawa structure, is the downstream target. It requires the simultaneous resolution of Deliverables 1 and 2 above, and closure of $[\text{H-color}]_{\text{pointwise}}$.

9 Conclusion

The fermionic matter sub-programme establishes that fermions, chirality, hypercharge, and three generations are not external additions to the Cosmochrony framework. They are the spinorial face of the admissible Weil module V_ρ, revealed through the Lorentzian complexification of its metaplectic structure.

The electroweak bundle $\mathrm{SU}(2)_L \times \mathrm{U}(1)_Y$ arises from the tensor functors of the admissible spinor bundle $\mathcal{S}_\Pi$ (Theorem A, unconditional). The $V - A$ chiral structure and hypercharge rigidity arise from the projective endomorphism E_Π (Theorem B, conditional on Hypothesis 3.5). Three generations arise from the spinorial multiplicity reading of $\sigma_c(n_3) = 3$ (Theorem C, unconditional). The colour-coupled quark sector follows by tensoring with V_{color}, conditional on [H-color].

The remaining open components are the derivation of Hypothesis 3.5 from A1–A4, the explicit construction of E_Π, and the derivation of the Yukawa sector and mass spectrum.

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